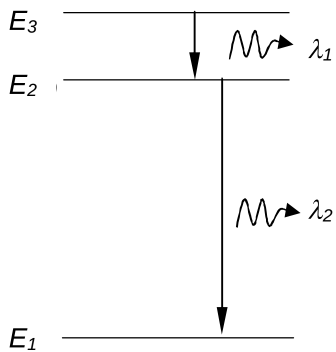
28 The diagram below shows a simplified representation of the three electron energy levels in an atom.



Cool vapour of this element at low pressure is bombarded with electrons accelerated from rest across a potential difference V. Two possible transitions which result in the emission of photons of wavelengths $\lambda_1 = 6.22 \times 10^{-7}$ m and $\lambda_2 = 1.78 \times 10^{-7}$ m are observed.

What is the minimum value of V for the above transitions to occur?

A 1.56 V **B** 2.80 V **C** 7.00 V **D** 9.00 V

29 A stationary uranium nucleus of mass 238 units disintegrates by the emission of an α particle